Progress Checkpoint 1

An AI-Assisted Blue Team Detection Agent for Windows Enterprise Environments

Capstone Group 1

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1. Current Project Status

Planned for this checkpoint. Milestone M1 (lab and telemetry baseline: VMs, audit policy, Sysmon, Wazuh agents, and an LLM access decision) and the start of M2 (a first vertical slice of Scenario 1).

Completed.

- Lab VMs built and running: a Wazuh server, two Windows Server 2022 targets (WinServer2022-CYBR4750a and -b), and a configured Kali attacker VM.
- Wazuh fully installed (manager, indexer, dashboard, filebeat all active); both Windows agents enrolled and Active.
- Sysmon v15.22 installed on both Windows servers with a validated configuration and generating events in the Operational channel.
- Windows Security Event 4625 (failed logon) confirmed to reach Wazuh, and detection rule R1 (rule IDs 100100 and 100101) written. R1 validation is in progress (Section 4).
- Red Team Execution Plan written (attack steps for all three scenarios, the 15-run test procedure, evidence log fields, and a week-by-week schedule).
- Local LLM option (Ollama, llama3.1:8b) installed on the Wazuh server and tested on a real alert; performance problem identified (Section 6).

Schedule. Slightly behind schedule. The virtual machines repeatedly crashed during setup, which cost time and delayed the first end-to-end slice. Now that the machines are up and running, progress has picked up. The AI triage component is the main schedule risk.

Changes since the Design Review Package. The lab runs on a Proxmox/QEMU host rather than VirtualBox, VMware, or Hyper-V, with VMs named WinServer2022-CYBR4750a/b on the 10.46.254.x network. Following Professor Crandall feedback, we are prioritizing one complete vertical slice (Scenario 1) before expanding, and plan to reduce formal testing (see Section 7).

2. Environment Setup

Component	Status	Evidence
Wazuh server (Ubuntu, wazuh-server)	**Working.** Manager, indexer, dashboard and filebeat report active; ports 443, 1514, 55000 and 9200 listening.	Figure 4
WinServer2022-CYBR4750a (10.46.254.110) and -b (10.46.254.111)	**Working.** Windows Server 2022 evaluation; both Wazuh agents (v4.14.9) show Active.	Figure 5
Sysmon v15.22	**Working** on both servers. Service running; Event IDs 1, 4 and 22 observed.	Figures 1–3
Kali attacker VM	**Configured.**	n/a
Source control / file storage	**Partial.** Microsoft Teams is in use for files and coordination; the private GitHub repository planned in the design is not set up yet.	n/a
LLM access	**Partially working.** Ollama and llama3.1:8b installed on wazuh-server but too slow to use; fix in progress (Section 6). No external API budget confirmed.	Section 6

Still to configure: audit policy coverage for all required event IDs (4624, 4672, 4720, 4732, 5140, 5145, System 7045) and Sysmon Event ID 3, verification of telemetry from both hosts, snapshots for scenario resets, and a working LLM path. We currently have access to all required resources except a fast LLM option.

3. Initial Research and Technical Preparation

- Wazuh documentation guided the installation and the custom rule structure. R1 is built as two rules: a base rule matching failed network logons (100100) and a frequency rule that flags repeated failures from one source IP within 60 seconds (100101). Keeping each rule small and replayable follows the instructor's advice to prefer simple, explainable detections.
- Microsoft auditing and Sysmon documentation drove our decision to verify that each event actually appears in Wazuh before writing rules (our Risk 1 mitigation). Sysmon adds command-line and parent-process detail that Windows Security logs lack, which Scenarios 2 and 3 need.
- MITRE ATT&CK (T1110, T1021.001, T1078, T1136.001, T1021.002, T1569.002) defines what each rule must detect and gives the AI service a trusted technique label taken from rule metadata, not from the model.
- OWASP LLM01 (prompt injection) shaped the triage service design: log fields are treated as untrusted, delimited, length-limited, and validated against a fixed JSON schema, and the service has no write access to the lab.
- Ollama / llama.cpp behavior explained our slow inference: CPU-only inference depends on instruction-set features (such as AVX2) that the VM's generic CPU type hides, which led to the Proxmox change we are requesting.

4. Proof of Concept and Early Implementation

The monitoring pipeline works from endpoint to SIEM. Sysmon runs on the Windows servers and records process creation with full command lines; launching notepad.exe, for example, produced a Sysmon Event ID 1 showing the image, command line, user and parent details (Figure 3). Both Windows agents report Active in the Wazuh dashboard (Figure 5), and the Wazuh stack is healthy and listening on its agent and API ports (Figure 4).

For Scenario 1, we made three deliberate failed logons against WinServer2022-CYBR4750a and confirmed the full path: Windows logged three Event 4625 entries (logon type 3, NTLM, from 10.199.28.51; Figure 6), the agent is configured to collect the Security channel (Figure 9), and the events arrived in the Wazuh server's alerts.json, where Wazuh's built-in rule 60122 ("Logon Failure – Unknown user or bad password") fired for each one (Figure 7). We then wrote our R1 rules (Figure 8): 100100 matches a failed network logon (logon type 3 or 10) as a child of 60122, and 100101 fires on 5 failures from the same source IP within 60 seconds and carries the ATT&CK tag T1110. We backed up local_rules.xml first (Figure 10). Because only three failures were generated, the 100101 threshold has not yet been exercised; the next test is five or more failures from Kali after reloading the rules.

5. Technical Evidence

Fig.	Artifact	What it shows and why it matters
1, 2	Sysmon installation and service check (command output)	Sysmon v15.22 installed with a validated config; `sc query` shows Sysmon64 RUNNING. This is the process-creation telemetry source for rules R4 and R6.
3	Event Viewer, Sysmon/Operational	Live Sysmon events with command-line detail, showing the telemetry exists before rule writing.
4	Wazuh services and listening ports	The full Wazuh stack is healthy, with agent port 1514 ready.
5	Wazuh Endpoints page	Both Windows servers enrolled and Active, supporting success criterion SC-1.
6	Event 4625 detail (PowerShell on the target)	Windows logs the failed logon with logon type, source IP and account: the raw data rule R1 depends on.

Fig.	Artifact	What it shows and why it matters
7	Wazuh alerts.json, rule 60122	Events travel from the Windows agent to Wazuh and trigger an alert with the full 4625 payload, proving agent-to-SIEM collection.
8	local_rules.xml with R1	The team's first custom detection logic: rules 100100 and 100101, tagged to ATT&CK T1110.
9, 10	Agent ossec.conf; rules backup	Agent collects the Security channel; the rules file was backed up before editing.

Screenshots appear in the Appendix.

6. Challenges and Blockers

Challenge 1: Virtual machine instability during setup.

- **What happened:** the VMs repeatedly crashed, we ran short of storage space, and the GUI caused additional problems.
- **Effect:** it consumed time planned for rule development and delayed the first vertical slice.
- **Response:** we added space and resolved the GUI issues; the VMs are now configured and running, and Chad had to rebuild some VMs.
- **Help needed:** Had multiple meetings with Chad.

Challenge 2: LLM inference is too slow.

- **What happened:** Ollama with llama3.1:8b on wazuh-server timed out twice (over 10 minutes) on a real alert. The VM's CPU type, "QEMU Virtual CPU 2.5+", hides the instruction-set features the model needs.
- **Effect:** it blocks the alert-to-summary step of the vertical slice.
- **Response:** we are working with the Chad to change the CPU type to host, add cores (target 8), and power-cycle the VM, and we will confirm the cause by checking CPU flags with `lscpu` before and after. In parallel we plan to shorten prompts, cap output length, and test a smaller (about 3B) model as a fallback.
- **Help needed:** the Proxmox change, which is in progress.

Finding: the default agent filter hides Event 5145.

- **What happened:** while checking the agent configuration (Figure 9), we found the default Wazuh Windows agent query excludes Event IDs 5145, 5156 and 5447.
- **Effect:** rule R5 (admin-share access) depends on 5145, so Scenario 3 would have been partly invisible.
- **Response:** we caught it before writing R5 and will remove the exclusion and re-verify telemetry (Next Step 2).
- **Help needed:** none.

7. Scope Review

- **Is the required scope still realistic?** Yes. The main risk is the AI triage step (LLM speed and access), not the lab or detection work.
- **Removed, reduced, or changed?** Under consideration, per instructor feedback: three successful runs per scenario instead of five, a shorter benign activity test instead of the 48-hour baseline, and a small prompt-injection test. Rules will favor simple, reliable detections over complex correlation. We aim to complete at least five of the six planned rules and will discuss any shortfall with the professor. Changed: the lab runs on Proxmox/QEMU, and files are kept in Teams until the GitHub repository is created.
- **Stretch goals in the required scope?** None.
- **Moved to future work?** Nothing new. Cloud migration, Active Directory, and automated response remain future scope, and the web view stays a stretch goal. If the local model stays too slow, we will use a smaller local model or a hosted API through our provider-neutral interface.

8. Team Contributions

Team Member	Contributions Since the Design Review
Michael Tesfaye	Installed Ollama and llama3.1:8b on wazuh-server, tested it on a real alert, diagnosed the slow inference cause, and prepared the Proxmox change request. The Python triage service code has not been started yet.
Laurent Ortiz	Configured Sysmon and Wazuh on the VMs with Sidney, helped set up the environment, started writing the detection rules (R1: 100100 and 100101), backed up the Wazuh rule files, and verified Event 4625 collection.
Sidney Nunez	Configured Sysmon and Wazuh on the VMs with Laurent and made sure the lab environment was set up and running, including getting the VMs stable after repeated crashes.
Ashton Eastman	Wrote the Red Team Execution Plan: attack steps and ATT&CK mapping for all three scenarios, the 15-run test procedure, evidence and ground-truth logging, and a week-by-week schedule. Kali VM configured.

9. Next Steps (before Progress Checkpoint 2)

1. Apply the Proxmox CPU change, then show `lscpu` flags before and after and one real alert summarized without timing out, with the response time recorded (the full demo cycle must fit in 10 minutes). If it is still too slow, switch to the fallback model or API.
2. Remove the Event 5145 exclusion from the agent query, then verify that Events 4624, 4672, 4720, 4732, 5140, 5145, System 7045, and Sysmon Event IDs 1 and 3 all arrive in Wazuh from both Windows hosts.
3. Reload the Wazuh rules, confirm 100100 and 100101 fire (using wazuh-logtest and live events) on 5 or more failed RDP logons from Kali within 60 seconds, and complete rule R2.
4. Run Scenario 1 end to end (Kali RDP guessing, 4625/4624, Wazuh alert, alerts.json) and record it in the run log.
5. Start the Python triage service so it reads a Wazuh alert from alerts.json and returns a schema-validated JSON summary, and create the GitHub repository for rules and code.
6. Have a second team member repeat Scenario 1 to complete the vertical slice.

Appendix: Technical Evidence

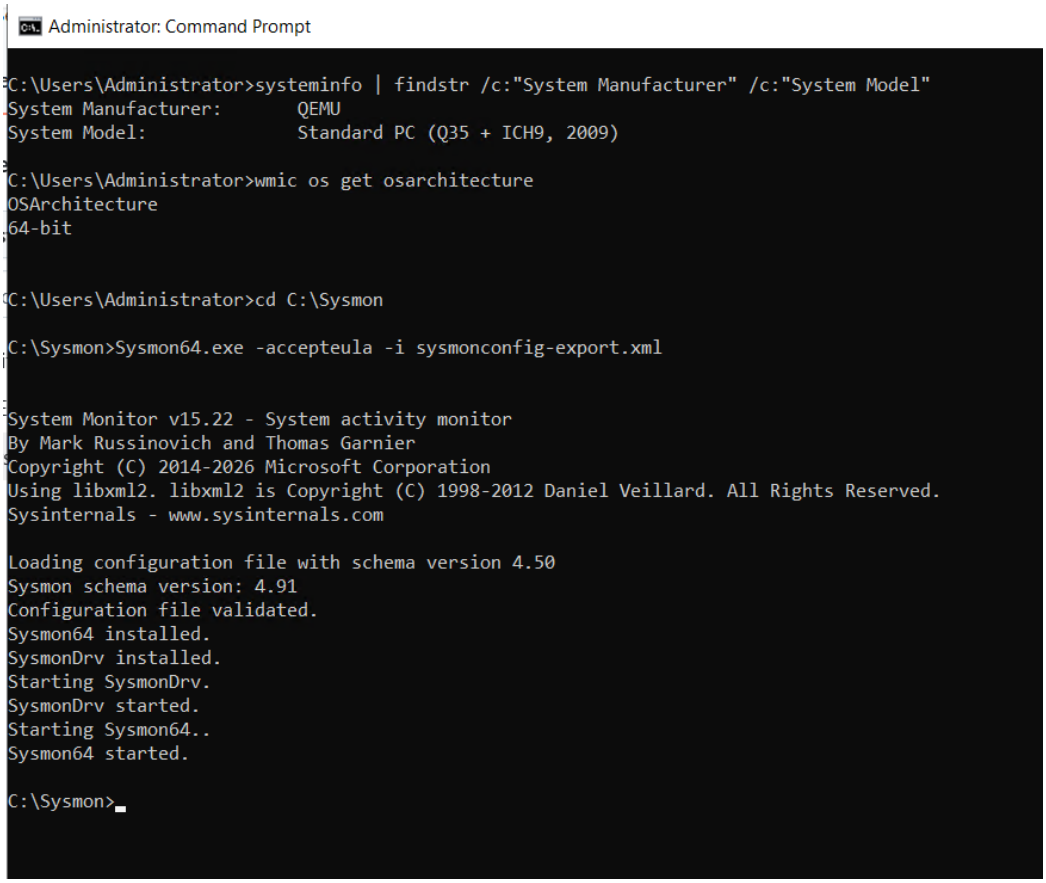


Figure 1. *Sysmon64 installed from the command line (Server 2022 on QEMU, 64-bit). Shows the Sysmon driver and service installed and started with a validated configuration file.*

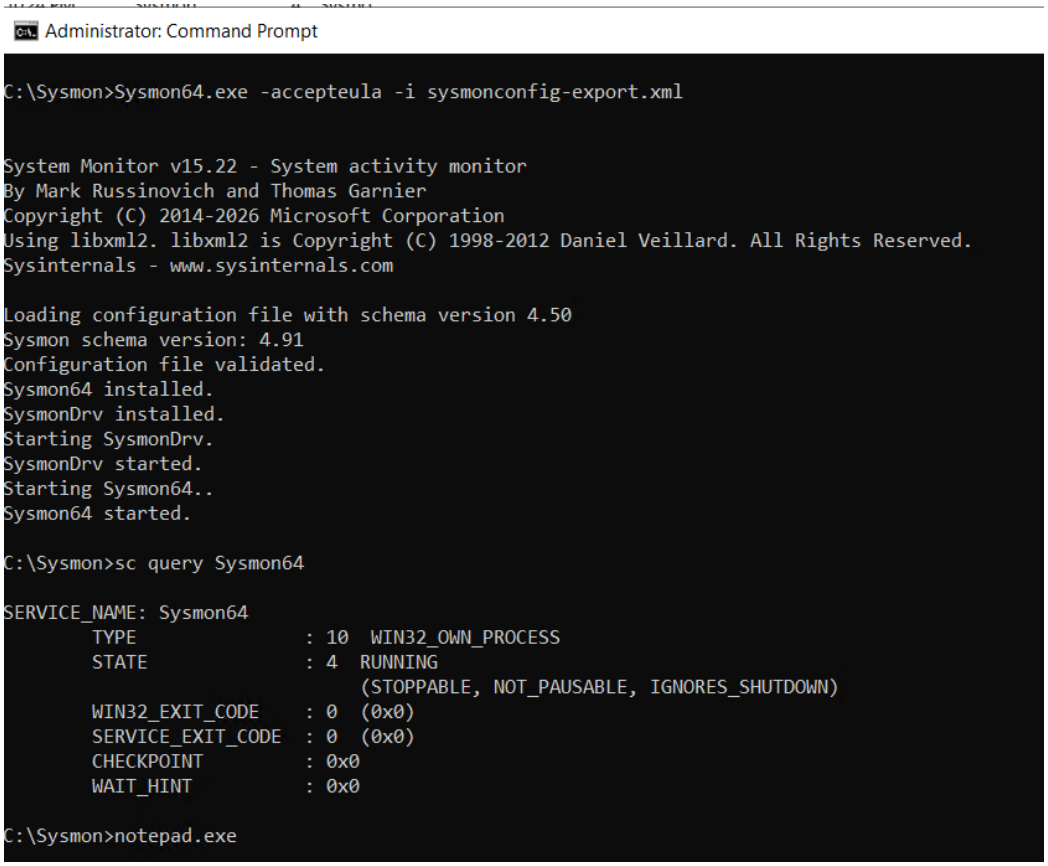


Figure 2. *Sysmon service verified. `sc query Sysmon64` shows STATE: RUNNING, confirming persistent process-creation telemetry.*

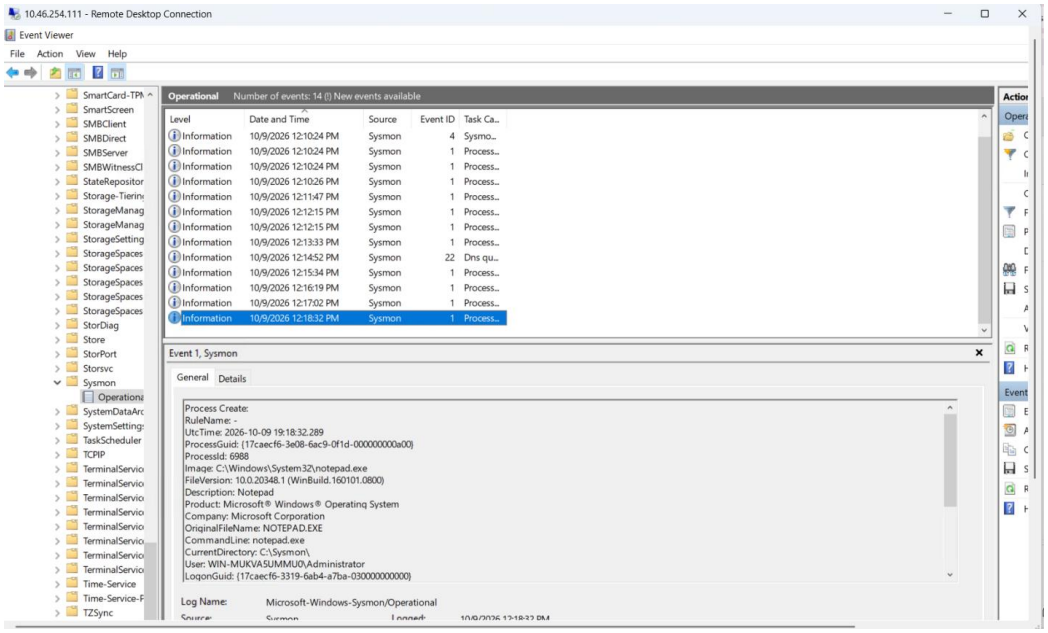


Figure 3. *Event Viewer, Microsoft-Windows-Sysmon/Operational on 10.46.254.111. Event ID 1 for notepad.exe shows image, command line, user and parent process, the detail needed by rules R4 and R6.*

```
root@wazuh-server:~# sudo systemctl is-active wazuh-manager wazuh-indexer wazuh-
dashboard filebeat
active
active
active
active
root@wazuh-server:~# sudo ss -tlnp | grep -E ':443|:1514|:55000|:9200'
LISTEN 0      2048             0.0.0.0:55000      0.0.0.0:*    users:(("python3
",pid=1202,fd=47))
LISTEN 0      511              0.0.0.0:443        0.0.0.0:*    users:(("node",p
id=516,fd=19))
LISTEN 0      128              0.0.0.0:1514       0.0.0.0:*    users:(("wazuh-r
emoted",pid=1504,fd=4))
LISTEN 0      2048                [::]:55000         [::]:*    users:(("python3
",pid=1202,fd=48))
LISTEN 0      4096   [::ffff:127.0.0.1]:9200            *:*    users:(("java",p
id=654,fd=630))
root@wazuh-server:~#
```

Figure 4. *Wazuh server health. wazuh-manager, indexer, dashboard and filebeat are active; ports 443, 1514 (agents), 55000 (API) and 9200 are listening.*

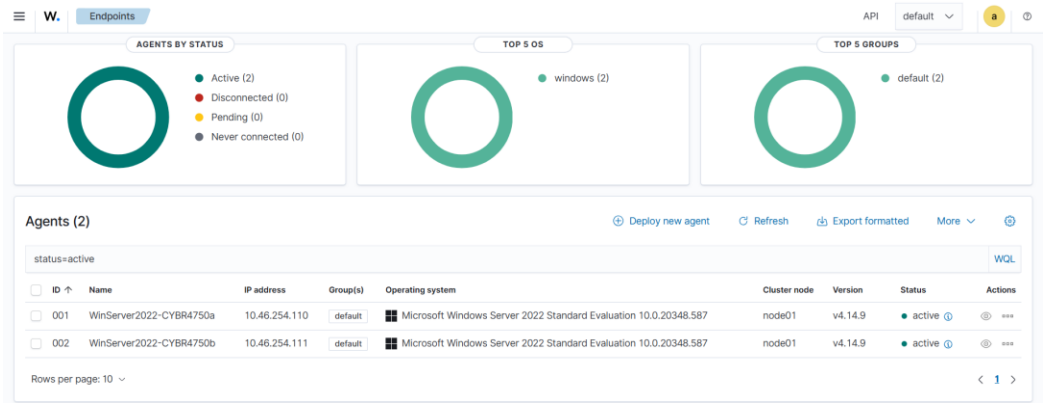


Figure 5. *Wazuh dashboard, Endpoints. Both Windows Server 2022 agents (001, 002) are Active, with 0 disconnected.*

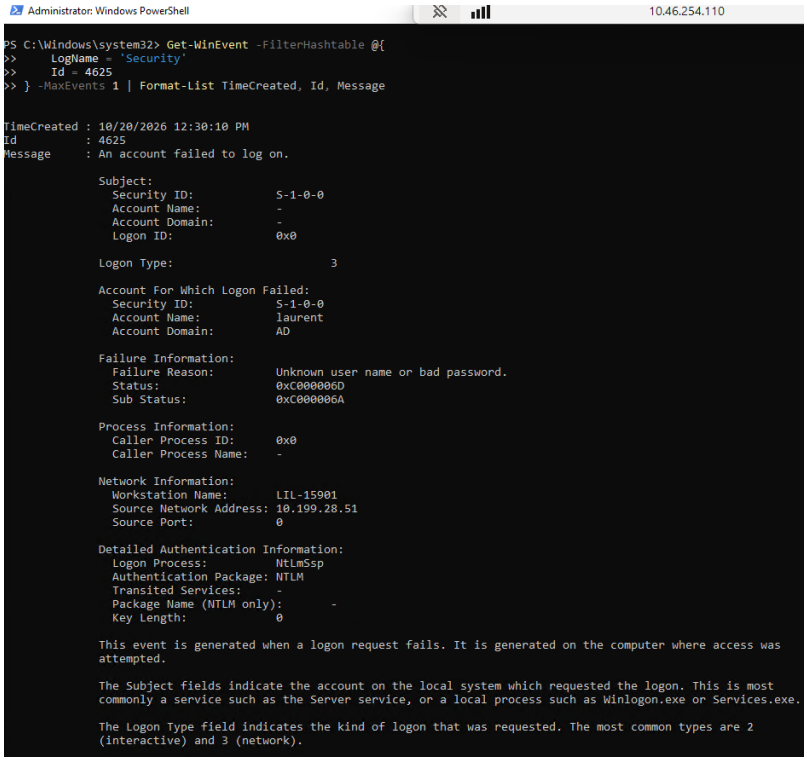


Figure 6. *Event 4625 detail on 10.46.254.110. Logon type 3 (network), NTLM, account "laurent", source address 10.199.28.51: the fields rule R1 matches on.*

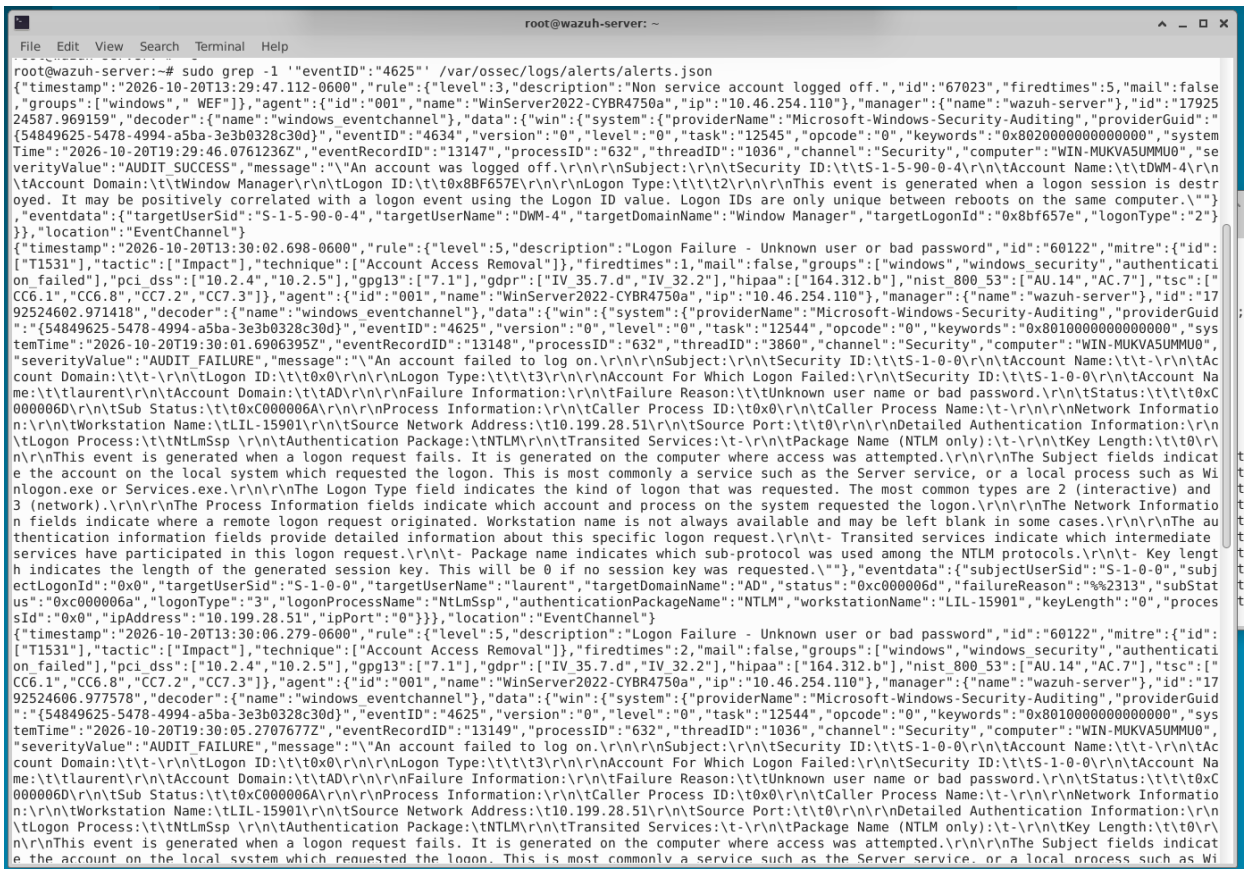


Figure 7. *alerts.json on wazuh-server. The three 4625 failures arrive from agent 001 (WinServer2022-CYBR4750a) and trigger built-in rule 60122 with the full event payload.*

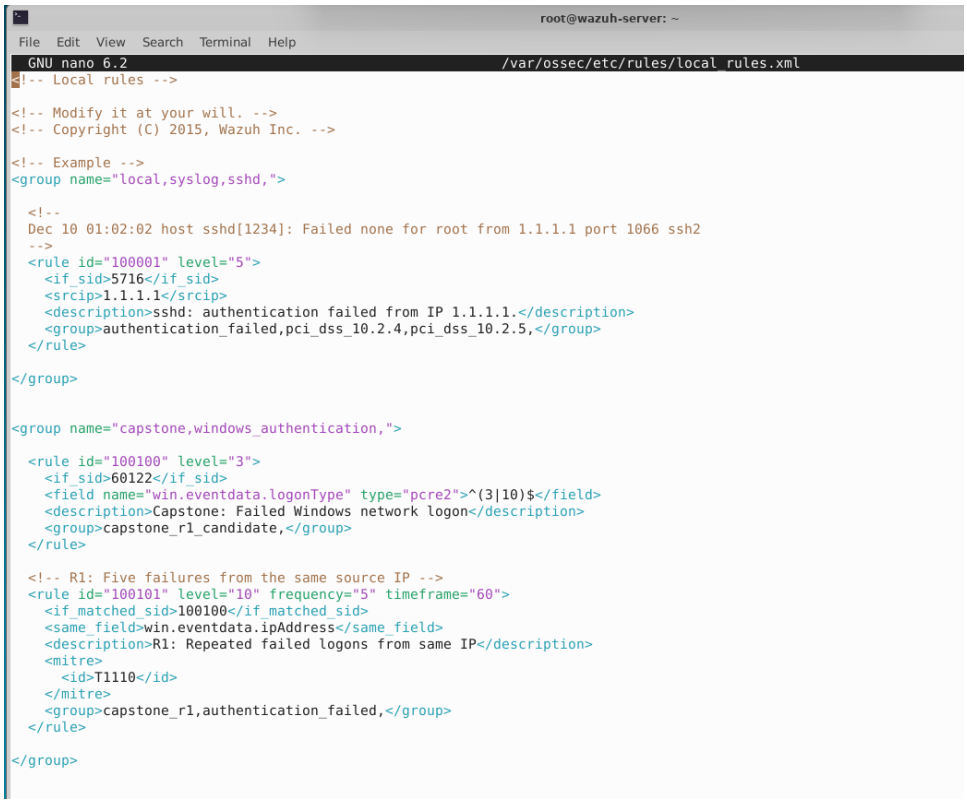


Figure 8. *local_rules.xml. Rule 100100 (failed network logon, child of 60122) and rule 100101 (frequency 5 in 60 seconds, same source IP, ATT&CK T1110).*

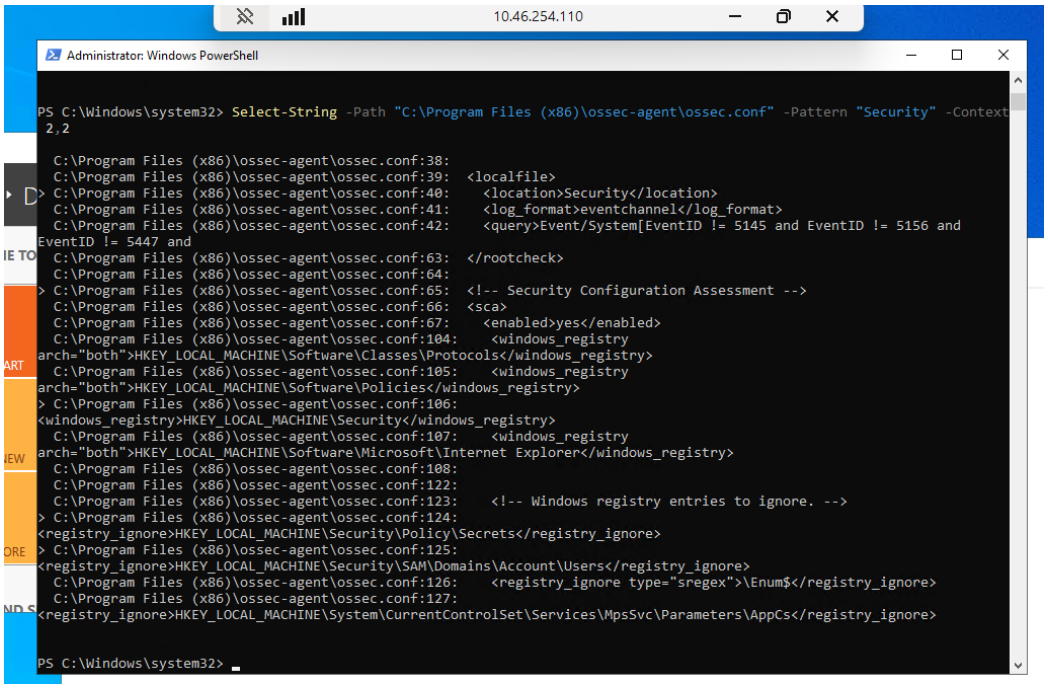


Figure 9. *Agent ossec.conf on the Windows target. The Security channel is collected; note the default query excludes Event IDs 5145, 5156 and 5447.*

```
root@wazuh-server:~# sudo ls -l /var/ossec/etc/rules/local_rules.xml
-rw-rw---- 1 wazuh wazuh 497 Oct  8 07:54 /var/ossec/etc/rules/local_rules.xml
root@wazuh-server:~# sudo ls -ld /var/ossec/etc/rules
drwxrwx--- 2 root wazuh 4096 Oct 15 16:42 /var/ossec/etc/rules
root@wazuh-server:~# sudo cp /var/ossec/etc/rules/local_rules.xml /var/ossec/etc/rules/local_rules.xml.bak
root@wazuh-server:~# sudo ls -l /var/ossec/etc/rules
total 8
-rw-rw---- 1 wazuh wazuh 497 Oct  8 07:54 local_rules.xml
-rw-r----- 1 root  root  497 Oct 20 14:11 local_rules.xml.bak
root@wazuh-server:~#
```

Figure 10. *Backup of local_rules.xml created on wazuh-server before rule changes.*